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Demonstration question paper generated from available previous papers - for practice only

B.Tech. II Semester Examination, 2026

Regulation: R20 | University: JNTUK

Machine Learning (21CS61)

Time: 3 hours Max. Marks: 60

Answer one question from each unit. All questions carry equal marks.

UNIT - 1 [Introduction & Regression]

- a) Explain gradient descent and its variants.
- b) Derive the normal equation for linear regression.

UNIT - 2 [Classification & Model Evaluation]

- a) Explain logistic regression with the sigmoid function.
- b) Explain the SVM algorithm with the kernel trick.

UNIT - 3 [Unsupervised Learning & Dimensionality Reduction]

- a) Explain k means clustering with an example.
- b) Explain PCA and its role in dimensionality reduction.

UNIT - 4 [Neural Networks & Deep Learning]

- a) Explain backpropagation with a neat diagram.
- b) Explain backpropagation with a neat diagram.

UNIT - 5 [Ensemble Methods & Model Selection]

- a) Compare bagging and boosting techniques.
- b) Compare bagging and boosting techniques.

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